

CHAPTER 7

Technical SEO (Complete Masterclass)

Learning Objectives

By the end of this chapter, you will be able to:

- Understand Technical SEO from scratch.
 - Learn how Google crawls and indexes websites.
 - Understand website architecture.
 - Learn XML Sitemap, Robots.txt, Canonical Tags.
 - Understand HTTPS.
 - Learn Mobile-First Indexing.
 - Understand Core Web Vitals.
 - Learn Structured Data (Schema Markup).
 - Understand redirects.
 - Fix common Technical SEO issues.
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What is Technical SEO?

Definition

Technical SEO is the process of optimizing the **technical aspects** of a website so that search engines can efficiently **crawl, index, understand, and rank** its pages.

Unlike On-Page SEO (content) and Off-Page SEO (authority), Technical SEO focuses on the website's infrastructure.

Goal of Technical SEO

Technical SEO aims to ensure that:

- Search engines can access your website.
- Pages can be indexed.
- Users experience fast loading times.
- The site works well on mobile devices.
- The website is secure.
- Duplicate content issues are minimized.

Why is Technical SEO Important?

Imagine writing an excellent book but locking it inside a room that nobody can enter.

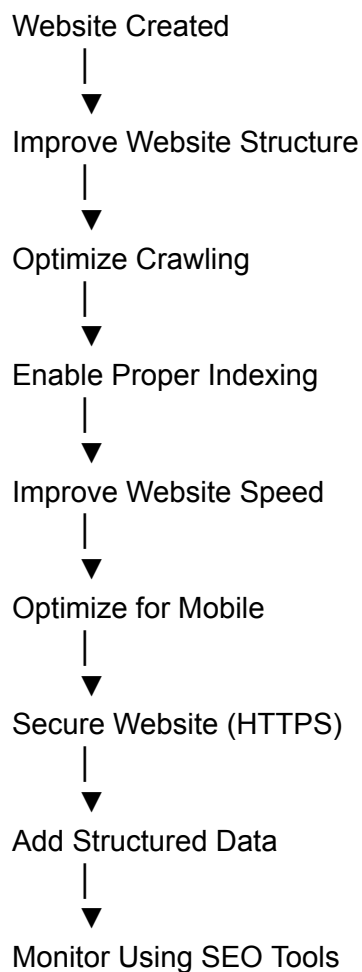
Even though the content is valuable, nobody—including Google—can access it.

Similarly, if a website has technical problems such as:

- Broken pages
- Slow loading
- Blocked crawlers
- Missing sitemap

its content may never reach users.

Technical SEO Workflow



Topic 1: Website Architecture

Definition

Website architecture is the way webpages are **organized and connected**.

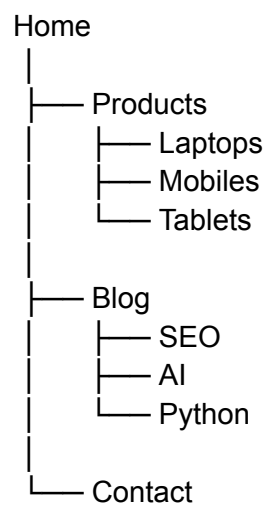
A good structure helps both:

- Users
- Search engines

navigate the website easily.

Example

Good Website Structure



Poor Website Structure

Home

↓

Page

↓

Another Page

↓

Another Page

↓

Random Page

↓

Product

Why Good Architecture Matters

A clear structure:

- Improves navigation.
 - Helps search engines discover pages.
 - Makes internal linking easier.
 - Improves user experience.
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Topic 2: Crawlability

Definition

Crawlability is the ability of search engine crawlers (such as Googlebot) to access and explore your website.

If Googlebot cannot crawl a page, it cannot index or rank it.

Factors Affecting Crawlability

- Broken links
 - Robots.txt restrictions
 - Server errors
 - Poor internal linking
 - Infinite URL loops
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Example

Imagine a supermarket.

If all aisles are open, customers can explore every product.

If some aisles are locked, customers cannot access those products.

Googlebot behaves similarly.

Topic 3: Crawl Budget

Definition

A **crawl budget** is the approximate number of pages a search engine chooses to crawl on your website within a certain period.

Large websites benefit from optimizing crawl budget so that important pages are crawled more frequently.

Factors That Influence Crawl Budget

- Website size
 - Server performance
 - Number of broken links
 - Duplicate pages
 - Internal linking
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Topic 4: Indexability

Definition

Indexability refers to whether a webpage can be stored in Google's index.

Only indexed pages can appear in search results.

Reasons a Page May Not Be Indexed

- `noindex` meta tag
- Blocked by `robots.txt`

- Duplicate content
 - Poor-quality content
 - Server errors
-

Topic 5: XML Sitemap

What is an XML Sitemap?

An **XML Sitemap** is a file that lists the important pages of your website.

It acts like a roadmap for search engines.

Example

```
<url>
  <loc>https://example.com/</loc>
</url>

<url>
  <loc>https://example.com/blog</loc>
</url>

<url>
  <loc>https://example.com/contact</loc>
</url>
```

Why is it Important?

It helps search engines:

- Discover new pages.
- Find updated pages.
- Understand website structure.

It is especially useful for:

- Large websites
 - New websites
 - Websites with many pages
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Topic 6: Robots.txt

Definition

The **robots.txt** file tells search engine crawlers which parts of the website they may or may not crawl.

Example

User-agent: *

Disallow: /admin/

Allow: /

This tells crawlers:

- Do not crawl the `/admin/` section.
 - Crawl the rest of the website.
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Important Note

Robots.txt does not automatically prevent indexing.

It controls **crawling**, not necessarily **indexing**. A page can still appear in search results in some situations if other signals point to it. To prevent indexing, the `noindex` directive (where supported) or other appropriate methods should be used.

Topic 7: Canonical Tag

Problem

Suppose these URLs all display the same content:

example.com/product

example.com/product/

example.com/product?ref=abc

Google may treat them as duplicate pages.

Solution

Use a **canonical tag** to tell search engines which version is the preferred one.

Example:

```
<link rel="canonical" href="https://example.com/product">
```

Benefits

- Reduces duplicate content issues.
 - Consolidates ranking signals.
 - Helps search engines understand the preferred URL.
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Topic 8: HTTPS

Definition

HTTPS is the secure version of HTTP.

It encrypts data exchanged between the user's browser and the website.

Benefits

- Protects user information.
 - Builds trust.
 - Improves security.
 - Is recommended for all websites.
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Topic 9: Mobile-First Indexing

Google primarily uses the **mobile version** of a website for indexing and ranking.

Why?

Most users now browse the web using smartphones.

Best Practices

- Responsive design.
 - Easy-to-read text.
 - Fast loading.
 - Clickable buttons.
 - Mobile-friendly navigation.
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Topic 10: Core Web Vitals

Core Web Vitals are performance metrics that measure user experience.

Largest Contentful Paint (LCP)

Measures how quickly the main content loads.

Goal:

≤ 2.5 seconds

Interaction to Next Paint (INP)

Measures how responsive a page is after user interactions (Google replaced the older First Input Delay metric with INP).

Lower values indicate better responsiveness.

Cumulative Layout Shift (CLS)

Measures visual stability.

A low CLS means elements do not unexpectedly move while the page is loading.

Topic 11: Structured Data (Schema Markup)

Definition

Structured Data is code added to a webpage to help search engines better understand its content.

Example

For a recipe page, structured data can specify:

- Recipe name
- Ingredients
- Cooking time
- Rating

This may help search engines display **rich results**.

Common Types

- Article
 - Product
 - FAQ
 - Recipe
 - Event
 - Organization
 - Local Business
 - Breadcrumb
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Topic 12: Redirects

What is a Redirect?

A redirect automatically sends users from one URL to another.

301 Redirect

Permanent redirect.

Use when a page has permanently moved.

302 Redirect

Temporary redirect.

Use when the move is only temporary.

Topic 13: Broken Links

Broken links lead to pages that no longer exist.

Example:

404 Page Not Found

Why Are Broken Links Bad?

- Poor user experience.
- Waste crawl resources.
- May reduce trust.

Regularly check and fix broken links.

Topic 14: Duplicate Content

Duplicate content means the same or very similar content appears on multiple URLs.

Problems

- Confuses search engines.
 - Splits ranking signals.
 - Makes it difficult to determine the preferred page.
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Solutions

- Use canonical tags.
 - Create original content.
 - Redirect duplicate pages where appropriate.
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Topic 15: Technical SEO Tools

Popular tools include:

Tool	Purpose
Google Search Console	Indexing, crawl errors, performance
Google PageSpeed Insights	Speed and Core Web Vitals
Screaming Frog	Crawl website and identify issues
GTmetrix	Website performance
Lighthouse	Performance, accessibility, SEO audits

Common Technical SEO Mistakes

- No XML Sitemap
- Incorrect Robots.txt configuration
- Slow website
- Broken links
- Missing HTTPS
- Duplicate content
- Poor mobile experience

- Too many redirect chains
 - Missing structured data
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Real-World Example

Imagine an online bookstore.

The content is excellent, but:

- Pages take 10 seconds to load.
- Half the product pages return 404 errors.
- The mobile version is difficult to use.

Even with good content, users may leave, and search engines may struggle to crawl and rank the site effectively.

After fixing these technical issues, the website becomes faster, easier to crawl, and more user-friendly.

Chapter Summary

Technical SEO focuses on the website's infrastructure.

Key concepts include:

- Website architecture
- Crawlability
- Crawl budget
- Indexability
- XML Sitemap
- Robots.txt
- Canonical tags
- HTTPS
- Mobile-First Indexing
- Core Web Vitals
- Structured data
- Redirects
- Broken links
- Duplicate content

A technically sound website provides a better experience for both users and search engines.

Interview Questions

1. What is Technical SEO?
 2. What is crawlability?
 3. What is crawl budget?
 4. What is an XML Sitemap?
 5. What is Robots.txt?
 6. What is a canonical tag?
 7. Why is HTTPS important?
 8. What is Mobile-First Indexing?
 9. What are Core Web Vitals?
 10. What is the difference between a 301 and a 302 redirect?
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Quick Revision Table

Concept	Purpose
Website Architecture	Organizes website structure
Crawlability	Allows search engines to access pages
Crawl Budget	Controls how many pages are crawled
Indexability	Determines whether pages can appear in search results
XML Sitemap	Lists important pages for search engines
Robots.txt	Guides crawler access
Canonical Tag	Identifies the preferred version of duplicate pages
HTTPS	Secures data transfer
Mobile-First Indexing	Uses the mobile version for indexing
Core Web Vitals	Measures user experience and performance
Structured Data	Helps search engines understand content
Redirects	Send users and crawlers to a different URL
Broken Links	Links to unavailable pages

Duplicate Content

Similar content on multiple URLs